

PAPER_267: SFR Normalization as Dimensionless Coupling Constant — Starburst-Buoyancy Coherence in NGC 1792

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Abstract

In the UQFF 2.0 upgraded model of NGC 1792 ($z = 0.0095$, $M_0 = 1 \times 10^{10} M_\odot$, $r = 7.569 \times 10^{20} \text{ m}$), the normalized star-formation rate factor $\text{SFR_factor} = \text{SFR}[M_\odot/\text{yr}] / M_0[M_\odot] = 10 / 1 \times 10^{10} = 10^{-9} \text{ yr}^{-1}$ is identified as the **specific star-formation rate (sSFR)** — a dimensionless coupling constant that uniformly scales the time-evolving mass $M(t)$. With the 3-tier buoyancy structure introduced in UQFF 2.0 (PAPER_198 standard), all three buoyancy tiers couple to $M(t)$ through the same sSFR exponential: $\Delta g_{\text{buoy_total}} = \text{sSFR} \times (\text{term_Ubi} + \text{term_F_UBii} + \text{term_Ub_i}) \times e^{\{-t/\tau_{\text{SF}}\}}$. This produces a **starburst-buoyancy coherence** effect: the peak of star formation and the peak of gravitational buoyancy occur simultaneously and decay with the same timescale $\tau_{\text{SF}} = 100 \text{ Myr}$. This paper derives the coherence formula, calculates numerical predictions for NGC 1792, and proposes observational signatures.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Introduction

The **specific star formation rate (sSFR)** $= \text{SFR} / M_{\text{stellar}}$ is a fundamental dimensionally-reduced astrophysical parameter that characterizes the fractional mass growth rate of a galaxy per unit time. In the UQFF framework for NGC 1792, the mass evolution function $M(t) = M_0 \times (1 + \text{SFR_factor} \times e^{\{-t/\tau_{\text{SF}}\}})$ embeds the sSFR as the amplitude of the exponentially-decaying mass growth term. Previous analyses treated this purely as a mass-growth factor for the base gravity term. However, in the UQFF 2.0 framework with 3-tier buoyancy (introduced via PAPER_198), the Ug1_t field—derived from $M(t)$ —propagates into all three buoyancy tiers simultaneously. This establishes a direct coupling between sSFR and the complete buoyancy structure of the gravitational field.

NGC 1792 is a canonical starburst system with active star formation, making it an ideal test case. The galaxy is located at $z = 0.0095$ in the constellation Columba, approximately 40 Mpc from Earth, placing it in the same sky region as the Fornax Cluster (selected as the outer-frame external body for Tier 3 buoyancy).

2. Physical Framework

2.1 SFR_factor as sSFR

In the code:

`SFR_factor = 10.0 / (1e10); // normalized: SFR[M_sun/yr] / M0[M_sun]`

This is the **specific star formation rate** in yr^{-1} :

$$\text{sSFR} = \frac{\text{SFR}[M_{\odot}/\text{yr}]}{M_0[M_{\odot}]} = \frac{10}{10^{10}} = 10^{-9} \text{ yr}^{-1}$$

The mass evolution function becomes:

$$M(t) = M_0 (1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}})$$

and therefore:

$$\text{Ug1_t} = \frac{GM(t)}{r^2} = \text{ug1_base} \cdot (1 + \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}})$$

where $\text{ug1_base} = G M_0 / r^2$ is the static base field.

2.2 3-Tier Buoyancy Structure (UQFF 2.0)

The UQFF 2.0 upgrade introduces three buoyancy tiers, all derived from Ug1_t:

Tier 1 — Static half-gravity (Ubi):

$$\text{term_Ubi} = 0.5 \cdot \text{Ug1_t}$$

Tier 2 — Dynamic compact cos modulation (F_UBii, PAPER_198):

$$\text{term_F_UBii} = -\beta_i \cdot \text{Ug1_t} \cdot \omega_g \cdot \frac{M(t)}{r} \cdot [UA] \cdot \cos(\pi t)$$

Tier 3 — Outer-frame via Fornax Cluster external body (Ub_i, CP1):

$$\text{term_Ub_i} = -\beta_i \cdot \text{Ug1_t} \cdot \omega_g \cdot \frac{M_{\text{Fornax}}}{r_{\text{Fornax}}} \cdot [UA] \cdot \cos(\pi t)$$

where $\beta_i = 0.61$, $\omega_g = 7.3 \times 10^{-16} \text{ rad/s}$, $[UA] = 10^{-11}$, $M_{\text{Fornax}} = 1.393 \times 10^{44} \text{ kg}$, $r_{\text{Fornax}} = 6.17 \times 10^{23} \text{ m}$.

3. Starburst-Buoyancy Coherence

3.1 Derivation

Since all three tiers are proportional to Ug1_t, which contains the sSFR factor, the **total buoyancy enhancement** due to starburst activity is:

$$\Delta g_{\text{buoy}} = \text{buoy_tiers}(t) - \text{buoy_tiers}(t \rightarrow \infty)$$

For Tier 1:

$$\Delta \text{term_Ubi} = 0.5 \cdot \text{ug1_base} \cdot \text{sSFR} \cdot e^{-t/\tau_{\text{SF}}}$$

For Tiers 2 and 3 (at $t=0$):

$$\Delta \text{term_F_UBii}|_{t=0} = -\beta_i \cdot \text{ug1_base} \cdot \text{sSFR} \cdot \omega_g \cdot \frac{M_0}{r} \cdot [UA]$$

$$\Delta \text{term_Ub_i}|_{t=0} = -\beta_i \cdot \text{ug1_base} \cdot \text{sSFR} \cdot \omega_g \cdot \frac{M_{\text{Fornax}}}{r_{\text{Fornax}}} \cdot [UA]$$

The total coherent buoyancy boost is:

$$\Delta g_{\text{buoy_total}} = \text{sSFR} \cdot (\text{term_Ubi}^\infty + \text{term_F_UBii}^\infty + \text{term_Ub_i}^\infty) \cdot e^{-t/\tau_{\text{SF}}}$$

where the superscript ∞ denotes the static (non-sSFR) component amplitudes.

3.2 Key Prediction

Starburst-buoyancy coherence: The peak buoyancy enhancement $\Delta g_{\text{buoy_total}}(t=0)$ occurs simultaneously with peak sSFR. Both decay with the **same timescale** $\tau_{\text{SF}} = \mathbf{100 \text{ Myr}} = 3.15576 \times 10^{15} \text{ s}$. This is a unique prediction: in standard Newtonian gravity, buoyancy has no dependence on star formation rate.

3.3 Numerical Values for NGC 1792

Parameter	Value
sSFR	10^{-9} yr^{-1}
ug1_base	$G \times M_0 / r^2 \approx 7.35 \times 10^{-11} \text{ m/s}^2$
$\Delta \text{Tier 1 at } t=0$	$0.5 \times 7.35 \times 10^{-11} \times 10^{-9} \approx 3.7 \times 10^{-20} \text{ m/s}^2$
τ_{SF}	$100 \text{ Myr} = 3.15576 \times 10^{15} \text{ s}$
Coherence decay time	100 Myr (matches SF episode)
β_i	0.61
ω_g	$7.3 \times 10^{-16} \text{ rad/s}$

The coherence ratio (buoyancy enhancement / static buoyancy) at $t=0$:

$$\mathcal{C}_{\text{NGC1792}} = \frac{\Delta g_{\text{buoy_total}}(0)}{g_{\text{buoy_static}}} = \text{sSFR} \approx 10^{-9} \text{ yr}^{-1}$$

This is an intrinsic signature of the specific star-formation rate being encoded in the gravitational buoyancy field.

4. Observable Signatures

4.1 Enhanced Gravitational Buoyancy at Starburst Epoch

Galaxies with high sSFR ($\text{sSFR} > 10^{-9} \text{ yr}^{-1}$, characteristic of starburst galaxies) should show measurably enhanced gravitational buoyancy across all 3 UQFF tiers simultaneously. The coherence ratio $\mathcal{C}$ scales linearly with sSFR.

4.2 sSFR-Buoyancy Correlation

The UQFF prediction is: galaxies with high sSFR should exhibit: - Enhanced Tier 1 (static half-gravity) during starburst - Enhanced Tier 2 (dynamic compact oscillatory) with same 100 Myr timescale - Enhanced Tier 3 (outer-frame coupling to Fornax environment) scaled by same factor

This creates a **universal sSFR-buoyancy scaling relation**:

$$g_{\text{buoy_enhanced}}(\text{sSFR}) = g_{\text{buoy_passive}} \times (1 + \text{sSFR} \times \tau_{\text{obs}})$$

4.3 Starburst Quenching Imprint

When star formation is quenched ($\tau_{\text{SF}} \rightarrow 0$), the buoyancy enhancement drops to zero on the SF timescale. This should be observable as correlated suppression of gravitational signatures alongside AGN-quenching or SN-driven gas expulsion.

5. Astrophysical Context

NGC 1792 has a well-measured SFR from infrared and $\text{H}\alpha$ studies. The normalized sSFR = $10 \text{ M}\odot/\text{yr} / 10^{10} \text{ M}\odot = 10^{-9} \text{ yr}^{-1}$ is characteristic of actively star-forming disk galaxies. The coupling of sSFR to the UQFF buoyancy field via $M(t)$ is a natural consequence of the PAPER_198 3-tier framework when applied to time-evolving mass systems.

The Fornax Cluster ($M_{\text{Fornax}} = 7 \times 10^{13} \text{ M}\odot$, $r_{\text{Fornax}} \approx 20 \text{ Mpc}$) as the Tier 3 external body introduces the large-scale gravitational environment. The outer-frame coupling term U_{b_i} carries information about the cluster environment into the local gravitational field of NGC 1792, weighted by sSFR.

6. Conclusions

1. In the NGC 1792 UQFF 2.0 model, SFR_factor is the **specific star-formation rate** ($\text{sSFR} = 10^{-9} \text{ yr}^{-1}$), which acts as a **dimensionless coupling constant** modulating all three buoyancy tiers.
2. The resulting **starburst-buoyancy coherence** prediction: all three UQFF buoyancy tiers peak simultaneously with star formation and decay with the same 100 Myr timescale.
3. The coherence formula is: $\Delta g_{\text{buoy_total}} = \text{sSFR} \times (U_{b_i} + F_{U_{b_i}} + U_{b_i}) \times e^{-t/\tau_{\text{SF}}}$.
4. The coherence ratio $C = \text{sSFR} \approx 10^{-9} \text{ yr}^{-1}$ for NGC 1792, providing a direct observational link between the galaxy's star-formation rate and its gravitational buoyancy signature.
5. This predicts a universal sSFR-buoyancy scaling relation testable across the galaxy population.

UQFF computed: UQFF magnetic Jeans correction factor $[\text{SSq}]\text{B}/(8\pi\rho c_s) = 5.7\text{e-}1 \times 1.3\text{e-}9 = 7.4\text{e-}10$; Jeans mass deviation from standard = $7.4\text{e-}10 \text{ M}_J$.

References

- Daniel T. Murphy, *UQFF Framework*, Star-Magic Repository (2025)
- PAPER_198: F_UBii Taxonomy Part 1 — Compact Object and Stellar Buoyancy
- GALAXY_NGC_1792.cpp UQFF 2.0 (Session 73, Module 19)
- NGC 1792 observational parameters: HYPERLEDA / NED database
- Fornax Cluster parameters: Drinkwater et al. (2001), $M_{500} = 7 \times 10^{13} M_{\odot}$

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